EECS 16A Foundations of Signals, Systems, & Information Processing
Fall 2025 Discussion 5A

1. DTFS Puzzle! [Adapted from EE 120 Discussion]

Consider a discrete-time signal $x : \mathbb{Z} \to \mathbb{R}$ having the following properties:

i. $x[n+4l] = x[n], \quad \forall l \in \mathbb{Z}$

ii. $\sum_{n=-1}^{2} x[n] = 2$

iii. $\sum_{n=-1}^{2} (-1)^n \cdot x[n] = 4$

iv. $\sum_{n=-1}^{2} x[n] \cos\left(\frac{\pi}{2}n\right) = \sum_{n=-1}^{2} x[n] \sin\left(\frac{\pi}{2}n\right) = 0.$

(a) Determine the complex exponential Fourier series coefficients X_{-1}, X_0, X_1, and X_2 for the signal x. From the coefficients, determine and provide a well-labeled plot for the signal x.

Answer:

The DTFS expansion is valid for any period of x, not just its fundamental period. We usually use the fundamental period since it has the least coefficients to compute, but in this case all we know is that x is 4-periodic, so we will compute 4 DTFS coefficients.

$$\begin{aligned} x : \mathbb{Z} \to \mathbb{R}, \quad x[n+4l] &= x[n], \forall l \in \mathbb{Z} \\ &\Rightarrow \quad p = 4/k, \omega_0 = k\pi/2 \text{ for some } k \in \{1,2,4\} \end{aligned}$$

We compute the DTFS of x assuming $k = 1$, i.e., $p = 4$.

For the following steps, we use the hints/properties to pattern match into the analysis equation. We do this by figuring out which frequency each summation corresponds to.

$$\begin{aligned} \sum_{n=-1}^{2} x[n] &= 2 \\ \Rightarrow \sum_{n=-1}^{2} x[n](1)^n &= 2 \\ \Rightarrow \omega_0 0 = 0 \Rightarrow \sum_{n=-1}^{2} x[n]e^{i(0)n} &= 2 \\ \Rightarrow 4X_0 &= 2 \\ \Rightarrow X_0 &= \frac{1}{2} \end{aligned}$$

We turned 1 into a complex exponential because we want to figure out which frequency it corresponds to. We know that $1 = e^{i(0)}$ which means that this summation must correspond to the 0th multiple of

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ω_0. Namely, the summation corresponds to the analysis equation for the 0th Fourier Coefficient, X_0. Then, we just divide by the fundamental period that we calcaulated earlier to determine the value of X_0. This method is repeated below.

We can also take note of the fact that X_0 is always just the average value of the signal over its period, which is $2/4 = 1/2$.

$$
\begin{aligned}
\sum_{n=-1}^{2}(-1)^n x[n] &= 4 \\
(-1)^n &= (e^{-i\pi})^n \\
&\Rightarrow \sum_{n=-1}^{2}(e^{-i\pi n})x[n] = 4 \\
&\Rightarrow \sum_{n=-1}^{2}(e^{-i(\frac{4\pi}{4})n})x[n] = 4 \\
&\Rightarrow \sum_{n=-1}^{2}(e^{-i(\frac{2\pi}{4})(2)n})x[n] = 4 \\
&\Rightarrow \omega_0 = \frac{2\pi}{4}, k = 2 \\
&\Rightarrow 4X_2 = 4 \\
&\Rightarrow X_2 = 1
\end{aligned}
$$

For this summation, we had to do a little more work since we needed to ensure that we get ω_0 in the exponent of the complex exponential so that we could correctly figure out the Fourier coefficient that this summation corresponds to, by pattern matching the analysis equation.

$$
\begin{aligned}
\sum_{n=-1}^{2} x[n]\cos\left(\frac{\pi}{2}n\right) &= \sum_{n=-1}^{2} x[n]\sin\left(\frac{\pi}{2}n\right) \\
&\Rightarrow \text{Since } p = 4, \text{ we have } \omega_0 = \frac{\pi}{2}. \text{ We can rewrite } X_1 \text{ as:} \\
X_1 &= \frac{1}{4}\sum_{n=-1}^{2} x[n]e^{-i\frac{\pi}{2}n} \\
&\Rightarrow \text{By Euler's formula, } e^{-i\frac{\pi}{2}n} = \cos\left(\frac{\pi}{2}n\right) - i\sin\left(\frac{\pi}{2}n\right) \\
X_1 &= \frac{1}{4}\sum_{n=-1}^{2} x[n]\left(\cos\left(\frac{\pi}{2}n\right) - i\sin\left(\frac{\pi}{2}n\right)\right) \\
&= \frac{1}{4}\sum_{n=-1}^{2} x[n]\cos\left(\frac{\pi}{2}n\right) - \frac{i}{4}\sum_{n=-1}^{2} x[n]\sin\left(\frac{\pi}{2}n\right) = 0 - i0 = 0
\end{aligned}
$$

In this case, the cosine and sine terms from the property have the same frequency as the fundamental frequency $\omega_0 = \frac{\pi}{2}$. Therefore, it was possible to simply apply Euler's identity to the complex exponential in the summation. However, note that this will not always be the case.

Finally, for X_{-1}:

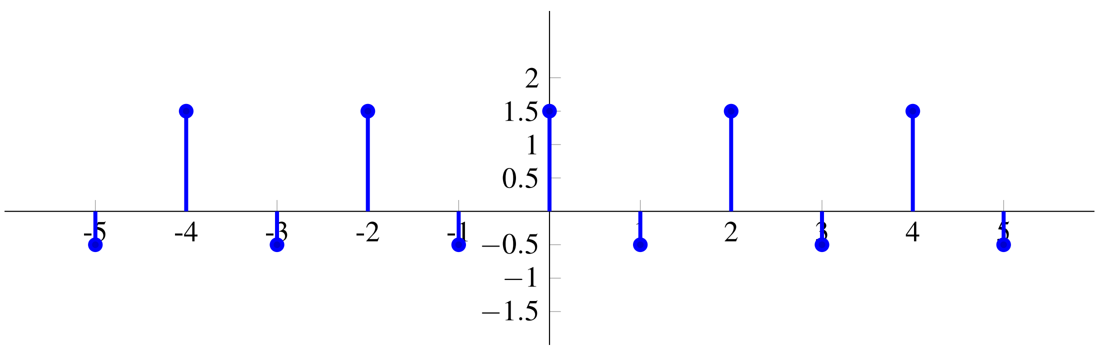
$$\begin{aligned} X_{-1} &= \frac{1}{4}\sum_{n=-1}^{2} x[n]e^{i\frac{\pi}{2}n} \\ &= \frac{1}{4}\sum_{n=-1}^{2} x[n]\left(\cos\left(\frac{\pi}{2}n\right) + i\sin\left(\frac{\pi}{2}n\right)\right) \\ &= \frac{1}{4}\sum_{n=-1}^{2} x[n]\cos\left(\frac{\pi}{2}n\right) + \frac{i}{4}\sum_{n=-1}^{2} x[n]\sin\left(\frac{\pi}{2}n\right) = 0 + i0 = 0 \end{aligned}$$

Notice that X_1 and X_{-1} exhibit symmetric behavior in the sense that the only difference in the analysis equation for X_1 and X_{-1} is that the sign in the exponent changes. So, the real part of both coefficients is the same while the imaginary part has a flipped sign. Namely, $X_1 = \overline{X}_{-1}$. **This is only true if x is a real signal.**

In conclusion,

$$X_{-1} = 0, \qquad X_0 = \frac{1}{2}, \qquad X_1 = 0, \qquad X_2 = 1$$

$$x[n] = \frac{1}{2} + e^{i\pi n}$$



(b) Based on your results from part (a), determine the fundamental period p and fundamental frequency ω_0 of the signal x. Express x in terms of its DTFS coefficients and complex exponentials of appropriate frequencies, and identify each coefficient and its corresponding harmonic frequency.

Answer:

From our result in part (a), we can see that $x(n)$ repeats every 2 samples and thus has a fundamental period (p) of 2. Its fundamental frequency is:

$$\omega_0 = \frac{2\pi}{p} = \pi.$$

Hence, from the DTFS synthesis equation,

$$x[n] = \sum_{k=0}^{1} X_k e^{j\frac{2\pi kn}{2}} = X_0 + X_1 e^{j\pi n}.$$

We've already seen that

$$x[n] = \frac{1}{2} + e^{j\pi n}$$

Hence, $X_0 = \frac{1}{2}$, which corresponds to frequency 0. $X_1 = 1$, which corresponds to frequency π radians. Note that in this specific question, x turns out to be 2-periodic, and can be expressed as:

$$x[n] = \frac{1}{2}e^{j0n} + e^{j\pi n}.$$

If we express $x[n]$ as a 2-periodic signal, then its 2 DTFS coefficients would be X_0, X_1 which are the coefficients of the exponentials $e^{j0n}, e^{j\pi n}$ respectively.

When we express x as a 4-periodic signal, we find that X_1, X_{-1} which correspond to $e^{j\frac{\pi}{2}n}, e^{j\frac{3\pi}{2}n}$ are 0, which makes sense, since we already determined that x only relies on $e^{j0n}, e^{j\pi n}$.

This holds generally as well - if we express a signal that is T-periodic as a (kT)-period signal with kT DTFS coefficients, then only the T coefficients corresponding to the T-periodic signal will be nonzero, and the other $(k-1)T$ coefficients will be 0.

Note: If we overestimate the fundamental period of the signal by an integer factor of 2, then the DTFS expression would include 4 coefficients, corresponding to the frequencies $0, \frac{\pi}{2}, \pi$, and $\frac{3\pi}{2}$. We would find:

$$x[n] = \sum_{k=0}^{3} X_k e^{j\frac{2\pi kn}{4}} = X_0 + X_1 e^{j\frac{\pi n}{2}} + X_2 e^{j\pi n} + X_3 e^{j\frac{3\pi n}{2}}.$$

In this representation, X_1 and X_3 would turn out to be zero, illustrating that they do not contribute to the signal periodicity since the true period is 2. The non-zero coefficients are X_0 and X_2, which correspond to e^{j0n} and $e^{j\pi n}$ respectively.

This holds generally as well—if we express a signal that is T-periodic as a (kT)-period signal with kT DTFS coefficients, then only the T coefficients corresponding to the T-periodic signal will be nonzero, and the other $(k-1)T$ coefficients will be 0. This highlights how we can overestimate a fundamental period, resulting in redundant zero-frequency components in the DTFS representation.

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